

Macronutrients and Calories

Calories

A calorie is a unit used to describe the amount of energy supplied by food and the amount of energy the body uses.

Food supplies energy mainly through:

- carbohydrate;
- protein;
- fat;
- alcohol.

Approximate energy values are:

- carbohydrate: **4 kcal per gram**
- protein: **4 kcal per gram**
- fat: **9 kcal per gram**
- alcohol: **7 kcal per gram**

Energy Balance

Body weight is strongly influenced over time by the relationship between energy intake and energy expenditure.

Energy Surplus

When average calorie intake is consistently higher than average calorie expenditure, body mass tends to increase.

A calorie surplus is commonly used during a muscle-gain phase.

Energy Deficit

When average calorie intake is consistently lower than average calorie expenditure, body mass tends to decrease.

A calorie deficit is commonly used during a fat-loss phase.

Energy Maintenance

When average calorie intake and expenditure are approximately balanced over time, body weight tends to remain relatively stable.

Daily body weight can still fluctuate because of:

- water;
- sodium intake;
- carbohydrate intake;
- glycogen;

- bowel contents;
- menstrual-cycle changes;
- inflammation from training.

A one-day scale change should not automatically be interpreted as a change in body fat.

Protein

Protein provides amino acids used for:

- muscle tissue;
- enzymes;
- hormones;
- immune proteins;
- structural tissues;
- repair and recovery.

Exercise increases the importance of adequate protein because training stimulates muscle-protein turnover.

A broadly useful range for many healthy exercising adults is approximately:

1.2-2.0 g of protein per kg of body weight per day.

Many resistance-training adults commonly aim toward the upper portion of this range.

Example for a 70 kg exercising adult:

- 1.2 g/kg = about 84 g/day
- 1.6 g/kg = about 112 g/day
- 2.0 g/kg = about 140 g/day

Protein Distribution

It is practical to distribute protein across several meals instead of consuming most of the day's protein in one meal.

Common protein-rich foods include:

- eggs;
- milk;
- curd or yogurt;
- paneer;
- fish;
- chicken;
- lean meats;
- whey protein;

- lentils;
- chickpeas;
- beans;
- soybeans;
- tofu;
- tempeh;
- peas;
- nuts and seeds.

Carbohydrates

Carbohydrates are an important source of energy.

They are particularly useful for:

- moderate- and high-intensity training;
- repeated exercise;
- endurance activity;
- replenishing muscle glycogen.

Carbohydrates are stored partly as glycogen in muscles and the liver.

Nutritious carbohydrate sources include:

- rice;
- roti;
- whole-grain breads;
- oats;
- potatoes;
- sweet potatoes;
- millet;
- fruits;
- vegetables;
- beans;
- lentils;
- dairy foods.

Carbohydrates are not inherently fattening. Body-fat gain is primarily associated with sustained excess energy intake.

Carbohydrate intake can temporarily increase body weight because stored glycogen is associated with water.

Dietary Fat

Dietary fat contributes to:

- hormone production;
- cell membranes;
- absorption of fat-soluble vitamins;
- essential fatty acid intake;
- energy intake;
- satiety.

Useful fat sources include:

- nuts;
- seeds;
- peanuts;
- olive oil;
- mustard oil;
- other vegetable oils;
- avocado;
- fatty fish.

Extremely low-fat diets are generally unnecessary for ordinary fitness goals.

Complete and Incomplete Proteins

A complete protein provides all essential amino acids in adequate amounts.

Many animal proteins and soy are complete proteins.

Some plant foods provide lower amounts of one or more essential amino acids, but plant proteins do not need to be combined at every single meal.

A varied diet across the day can supply the required amino acids.

Examples of complementary meal patterns include:

- rice with dal;
- roti with chickpeas;
- beans with grains.